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BhaskarOps

The Complete Guide: purpose, concepts, every feature, the maths behind the numbers, and what the app does not do

BhaskarOps (earlier name HeatOps) is a web platform that monitors urban heat across India and helps a district administration decide what to do about it: live readings for 1,932 cities, NASA satellite surface temperature, land cover, an AI-written Heat Action Protocol with its cost, and a budget planner.

This guide is written from the base level. It assumes no prior knowledge of meteorology, satellites or the code, and it explains each term the first time it is used.

Live site: https://heatops.vercel.app
Written from the code as deployed on 17 October 2026. Where a number on screen is an estimate, this guide says so, because the app says so.

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1. Why this app exists

Heat is the deadliest weather hazard in India and the least visible. A cyclone has a track and a landfall time; a heatwave arrives as a number on a thermometer that looks only slightly higher than yesterday's. Between 1992 and 2020 India recorded more than 25,000 heat-related deaths in official counts, and studies that measure "excess mortality" (how many more people died than usual) put the real figure far higher. The people who die are mostly those who cannot avoid the heat: construction workers, street vendors, delivery riders, farm labourers, the elderly in top-floor rooms, infants in tin-roofed housing.

Cities make it worse. Concrete, asphalt and tin absorb sunlight by day and release it at night, so a city centre can stay 3–6 °C hotter than the fields around it. This is the **urban heat island**. As India urbanises, more people live inside these islands every year.

India has a response framework. The National Disaster Management Authority (NDMA) publishes heatwave guidelines, the India Meteorological Department (IMD) issues warnings, and cities such as Ahmedabad have Heat Action Plans (HAPs): open cooling centres, move outdoor work to the morning, alert hospitals, send advisories. Ahmedabad's plan, started in 2013, is credited with avoiding over a thousand deaths a year.

The gap is operational. A district officer on a hot morning has the IMD bulletin and a phone, and has to answer: how bad is it here, today; which of the plan's steps should I trigger; what will it cost; and what happens if I do nothing. BhaskarOps is built to answer those four questions, for any of 1,932 Indian cities, from real data, in one screen.

2. What BhaskarOps is, and who it is for

BHASKAR stands for Bharat Heat Analysis, Surveillance, Knowledge & Assessment Resource; **OPS** for Optimization & Planning System. It is a website (React, runs in any modern browser) with a small server behind it for the AI calls and the data refresh.

Two audiences, two views

View	Who	What they see

Authority	District Magistrate, Municipal Commissioner, DDMA control room, health and labour officers, planners	The full platform: the Command Center map, the AI Co-Pilot with costs and consequences, the Mitigation Planner, satellite data, the city dashboard with all tabs
Citizen	Residents and families	A plain-language view: how hot it feels, a danger word, safe hours today, what to do, a WhatsApp share, and a simple chat with the AI

Authority is the default; the toggle is in the top bar. Everything in the citizen view is also in the authority view.

The one rule the whole app follows. A number that was not measured is labelled: "(estimated)", "illustrative", "model-based estimate", "no reading", or "no live data". The app never invents a reading for a city it has no data for, never estimates deaths, and never says "Agent37" or "satellite" about something that is not. When a judge asks "where does this number come from?", the answer is on the screen next to the number.

3. The concepts, explained from zero

These are the ideas the app is built on. Each is explained in plain terms first, then with the precise definition the app uses.

3.1 Air temperature versus how hot it feels

A thermometer in the shade, 2 metres above ground, gives the **air temperature**. This is what the weather report means by "36 °C". But the body does not feel air temperature alone; it feels how easily it can cool itself, and that depends mostly on **humidity** (the amount of water vapour in the air). Sweat cools you only when it evaporates, and it evaporates slowly in humid air. So 36 °C at 20 % humidity (Jaisalmer in May) feels very different from 36 °C at 70 % humidity (Kolkata in June).

The app shows three measures of "how hot it feels", each with a specific job:

Heat Index
The United States National Weather Service's "feels like" temperature. It combines air temperature and relative humidity into one number and has published danger bands: Caution (≥ 26.7 °C), Extreme Caution (≥ 32.2 °C), Danger (≥ 39.4 °C), Extreme Danger (≥ 51.7 °C). This is the number the app shows in big type as "Feels like X °C" with a danger word.

Wet-bulb temperature
The lowest temperature a wet surface (like sweating skin) can reach by evaporation. It is the physical limit of how much sweat can cool you. Laboratory studies (Vecellio et al., 2022) found that healthy young adults lose the ability to shed heat at a wet-bulb of about 31 °C; 35 °C is the theoretical survival limit even at rest in the shade. The app computes it with Stull's 2011 formula.

WBGT (Wet-Bulb Globe Temperature)
The international standard (ISO 7243) for deciding how long a person can work in heat. It blends the wet-bulb (humidity), air temperature and radiant heat from the sun. Occupational tables (the US Army's TB MED 507, used worldwide) give a work/rest ratio per hour for each WBGT band: for example at WBGT 31.1–32.2 °C a worker doing moderate work needs 30 minutes of rest every hour. The app uses this to write the labour order ("outdoor work 06:00–11:00") and to count work hours lost. No weather service reports radiant heat, so the app computes the *shade* WBGT and labels it a floor: in the sun the true value is higher.

3.2 Land surface temperature versus air temperature

A satellite cannot measure the air at 2 metres. It measures the temperature of the **ground surface** it sees: rooftops, roads, bare soil, fields. In sunshine the ground is much hotter than the air above it: a Jaipur rooftop can read 55 °C while the air is 42 °C, and bare rock in Leh can read 44 °C on a May morning while the air is 20 °C. This surface reading is called **LST** (land surface temperature).

Why use it at all? Because it is measured everywhere, at the same moment, by the same instrument, for twenty years. NASA's Terra satellite carries the MODIS instrument, which maps the whole Earth's surface temperature every day at 1 km resolution, passing over India at about 10:30 am (and 10:30 pm). The app holds MODIS readings for 1,912 cities since March 2026 and for 167 cities back to 2016. That is how the Historical slider can show 2016 against 2024, and how the planner knows a city's "season peak surface".

One catch: satellites cannot see through cloud. MODIS marks cloudy pixels as "no data", and the app keeps those as gaps rather than guessing. In the monsoon, most of India has no clear-sky reading for weeks; the app's "14-day composite" option shows each city's most recent clear-sky reading and says so.

3.3 The urban heat island

Built-up land absorbs more sunlight than vegetation and releases it slowly at night, traffic and air-conditioners add heat, and tall buildings block the wind. The result is a city that is warmer than its surroundings, most sharply at night. The app measures this directly: it fetches the live temperature at a point about 40 km outside the city (0.35° north, 0.10° east) and subtracts. A gap of more than 3.5 °C is labelled "Severe UHI effect".

3.4 Built-up fraction, vegetation, elevation

Built-up fraction is the share of the ground covered by man-made surfaces: buildings, roads, pavements, yards. It comes from the European Space Agency's WorldCover map (2021), which classifies every 10 m × 10 m square of the Earth from Sentinel satellite images. For each of 171 cities the app counts the classes in a box of about 5 km around the city centre: 58 % built-up for Raipur means 58 % of that box is sealed surface. It is land cover, not a count of buildings (the dashboard's Urban Morphology panel counts actual buildings from OpenStreetMap). Built-up share drives the heat island and sizes the planner's cool-roof packages.

Vegetation fraction is the share covered by trees, grass and crops. Trees cool their surroundings by shading and by *evapotranspiration* (releasing water vapour, which takes heat with it). The app compares a city's tree cover with the "3-30-300" target used by urban foresters (30 % canopy).

Elevation is height above sea level, from NASA's SRTM radar survey (30 m grid). Air cools by about 6.5 °C per 1,000 m of height, which is why Leh at 3,500 m is cold while Jaipur at 430 m is hot. The app uses elevation to pick the right IMD rule (see 3.5).

3.5 What counts as a heatwave: the IMD rules

IMD declares a heatwave when a station's maximum temperature reaches at least 40 °C in the plains, 37 °C on the coast, or 30 °C in the hills, *and* is either 4.5 °C above the normal for that date or reaches 45 °C outright (47 °C for a "severe" heatwave), on two consecutive days. The app applies the threshold and absolute rules to the 7-day forecast and calls the result a "threshold-based outlook, not an IMD warning", because it does not hold the per-city climatological normals the departure rule needs. Official warnings remain at mausam.imd.gov.in.

3.6 Air quality: AQI and PM

PM2.5 and **PM10** are particles in the air small enough to enter the lungs (2.5 and 10 micrometres across). The **AQI** (Air Quality Index) turns pollutant concentrations into one 0–500 number with health bands (Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, Hazardous). Heat and pollution compound: hot, still air traps dust and ozone, and the same people are vulnerable to both. The app uses the US EPA scale, delivered through Open-Meteo from the European CAMS model; it is not CPCB station data and the screen says so.

3.7 Interpolation: filling the space between cities

The app has readings at 1,932 points, but a map needs a value everywhere. **Inverse-distance weighting (IDW)** estimates the value at any location as an average of the nearest readings, with nearer readings weighted more (by $1/\text{distance}^2$). The Command Center uses IDW for the continuous thermal surface, for districts without their own city, and for the wind field. The rules that keep it honest: use only the nearest 8 readings, never reach farther than ~150 km on dense layers (~400 km on the sparse satellite and land-cover layers, with the result rounded and labelled "estimated from X"), and leave anything beyond that blank.

3.8 Choropleth, composite, semantic zoom

Choropleth
A map where each area (state, district) is filled with a colour that encodes a value. The app's state colour is decided by a vote: the risk bucket that the most cities in that state fall into (ties go to the hotter bucket, because a disaster map should not understate risk).

Composite
A satellite image built from several days, taking each location's most recent clear-sky reading. Standard practice for cloudy regions; the app labels the window ("last pass within 14 days").

Semantic zoom
The map shows different detail at different scales: a clean state view when zoomed out, per-district colours and borders when zoomed in or when a state is selected.

4. Where the data comes from

Data	Source	Coverage	Freshness
Air temperature, humidity, wind, forecast, UV	Open-Meteo forecast API (free, no key; aggregates national weather models and ECMWF)	1,932 cities in all 36 states and UTs	Bulk cache refreshed several times a day; the selected city fetched live
Air quality (US AQI, PM2.5, PM10, NO_2, O_3)	Open-Meteo air-quality API (Copernicus CAMS model)	Same	Same
Satellite land surface temperature	NASA MODIS MOD11A1 (Terra, 1 km), point samples via NASA AppEEARS; quality-filtered to clear-sky pixels	1,912 cities since 1 Mar 2026; 167 cities since 2016; daily national file of 184 dates	Static archive to 2 Sep 2026
Daily true-colour satellite image	NASA GIBS, VIIRS (Suomi NPP)	Whole country, one image per day	Previous day
Land cover (built-up, vegetation, water, bare)	ESA WorldCover 10 m, 2021	171 cities, ~5 km box each; other cities estimated from the nearest classified city, labelled	Static
Elevation	NASA SRTM 30 m via Open-Meteo	All located cities	Static
Building count	OpenStreetMap via Overpass	Selected city, 1 km radius	Live
Schools and hospitals	OpenStreetMap	6 cities, 3 km radius	Static
Satellite imagery basemap	Esri World Imagery (Maxar); Esri dark canvas	Whole world, to street level	Static
Map boundaries	GeoJSON: 36 states/UTs, 594 districts, official J&K/Ladakh file	India	Static
Daily archive	The app's own snapshots of the bulk cache	86+ days since 22 Jun 2026	One per refresh

The refresh pipeline

1. A script fetches all 1,932 cities from Open-Meteo in batches (temperature, cloud, rain chance, today's high, wind speed and direction, AQI, PM10) and writes `live-weather-cache.json` plus a dated snapshot.
2. GitHub Actions runs it hourly in the day and every three hours at night and commits the result, which redeploys the site automatically. A Vercel cron is the backup.
3. If a batch fails, those cities keep their previous values and are flagged "carried forward"; the map's status badge turns from LIVE to CACHED when the cache is older than 36 hours.

4. In the browser, selecting a city fetches its full live weather (7-day and hourly forecast, humidity, wind, UV); opening a state refreshes that state's cities live.

5. The screens: a guided tour

5.1 Sign-in

Register with any name, email and password (6+ characters), then sign in. Accounts live only in the browser's local storage; there is no server-side user database, which is deliberate for a demo and stated in the guide's limits. After sign-in an Authority user on a laptop lands on the Command Center; a Citizen, or any phone, lands on the classic map.

5.2 The Command Center (Authority, laptop)

A full-screen map with three floating panels and no page scrolling. It opens on an **All-India National Overview** with nothing selected: the left panel shows the hottest city right now, how many cities are forecast at or above 40 °C today, and the national mean; the right panel waits in standby. Clicking a state, clicking a district, or searching a city selects it and both panels switch to that locality. Section 6 covers every control.

5.3 The classic map

The original screen: the India map on the left, an information panel on the right with the national summary, quick picks (Delhi, Mumbai, Bengaluru, Jaipur, Chennai, Kolkata), and, when a state is selected, its city list with live temperatures, the state-level Mitigation Planner, and the Hottest Cities list. "Analyze" opens the city dashboard.

5.4 The city dashboard

Tabbed analysis of one city. Authority tabs: Overview, Prediction, Analysis, Compare, Interventions, AI + Export. Citizen tabs: Overview, What to do, Compare. Section 9 walks through each panel.

5.5 AGNI

The AI analyst, reachable from a floating button on every screen and from the AI + Export tab. Section 10.

5.6 Profile

Name, email, current view with a switch, points and badges, recent sign-ins.

6. The Command Center in detail

6.1 The map

The map is a proper tile-map engine (Leaflet), so it zooms from all of India (zoom 4) to individual city blocks (zoom 19) with the scroll wheel, the + / − buttons, or ◎ "fly to the selected city". The state and district boundaries are re-drawn as vectors at every zoom, so they stay crisp.

Base map: Dark	Esri dark canvas; streets and place names appear as you zoom in.
Base map: Satellite	Esri World Imagery (Maxar), the Google-Earth-style view, sharp to street level. From zoom 11, roads, rail and place labels are drawn on top.
Base map: NASA true colour	NASA's actual daily photograph of India (VIIRS). In Live Pass mode it follows the date you scrub, clouds included. Native resolution 250 m, so it is a weather-scale picture, not a street-scale one.

6.2 Layers: what the colours mean

Air temperature	Live Open-Meteo readings, 1,932 cities. In Forecast mode, today's forecast highs.
Surface temperature	NASA MODIS land surface temperature (ground, ~10:30 am). Latest clear-sky pass per city, a scrubbed day, or a year's April–June mean.
Built-up share	ESA WorldCover built-up percentage, 171 classified cities; others estimated from the nearest, labelled.

6.3 Colours and surfaces

Spectral thermal (default) is a continuous scale from violet (below 10 °C) through blue, cyan, green, yellow and orange to crimson (45 °C and above), so a 29 °C district and a 31 °C district are visibly different shades. **NDMA risk tiers** paints the six discrete buckets instead (Low < 25, Low-Moderate 25–30, Moderate 30–35, High 35–40, Very High 40–45, Extreme ≥ 45 °C). The tier *labels* appear in badges and tooltips under either palette. Built-up share has its own green (open land) → red (dense) scale with a legend.

District choropleth fills state and district polygons. **Continuous spectrum** replaces the fills with a smooth interpolated field computed from the city readings and clipped to India's outline, with the borders drawn over it, the way a meteorological chart looks.

6.4 Semantic zoom and per-district colours

Zoomed out, each state wears one colour (the bucket most of its cities fall in) and district borders are faint. The selected state is the exception: it shows its internal district-by-district gradient under a glowing cyan outline. From zoom 6, every district gets its own colour: 451 of the 594 districts share a name with a city and take that reading directly; the rest take a distance-weighted mean of the two nearest cities in their state (within 150 km); on the sparse satellite and land-cover layers the search widens to 400 km across state lines, rounded and labelled "estimated". Hovering a district shows its name, value, tier and source city; clicking it opens that city.

6.5 The timeline scrubber (bottom panel)

Historical	Drag through 2016–2026. Each year shows the April–June mean clear-sky surface temperature for the 166 cities with a multi-year satellite record, and the states recolour from them. The readout gives the selected city's figure for that year, including its hottest single day.
Live Pass	With Surface temperature on: drag day by day through 184 satellite passes (1 Mar – 2 Sep 2026) for the whole country. The caption gives the honest count: "149 cities with a pass that day, 1,110 shown from their last pass within 14 days, 653 none (cloud)". A checkbox switches the 14-day composite off to see exact-day coverage only. With NASA true colour on, the photograph follows the date.
Forecast	Drag 0–6 days: the readout shows the selected city's forecast high, low, condition and rain chance; the map shows today's forecast highs nationally.

6.6 Wind and heat advection

The "Live wind & heat advection flow" switch starts a Windy-style particle animation: 1,500 particles drift along the real 10-metre wind from the 1,932 city readings (interpolated between them), fading into trails. In **Thermal wind** mode each particle is coloured by the air temperature it carries, so hot westerlies crossing the plains show as orange and red streams. In **Stagnation zones** mode, places where the wind is under 6 km/h while the air is above 40 °C or PM10 above 150 µg/m³ pulse magenta: heat and dust the wind is not clearing. Hovering the map opens a probe with the wind vector ("24 km/h WNW (290°)"), an "advective heat load" reading and a "ventilation index"; both labels are heuristics and say so. Nothing is drawn farther than ~150 km from a city reading; no synthetic wind is ever added.

6.7 The left panel

Layer, base map, colour and surface switches; the wind controls; "Heat colours on the map" (off leaves borders only) and "Show city dots" (one dot per reading, off by default to keep the map clean); then, for the selected city, the **Alert triage** badge (temperature, activation level, feels-like, the reasons) and the 10 × 10 thermal grid, which starts from the city's real temperature and adds an illustrative ±2 °C intra-city pattern, labelled as such. With no city selected the panel shows the national summary.

6.8 The right panel

The AI Heat Response Co-Pilot (section 7) and the Heatwave Action Checklist (section 9.1).

6.9 The floating search

A translucent pill at the top of the map: type any of 1,932 city names; results show each city's live temperature; Enter picks the first; picking a city flies the map to it.

7. The AI Heat Response Co-Pilot

This is the feature that turns the platform from a monitor into an operations tool. It writes, for the selected city, a municipal Heat Action Protocol in the form a District Magistrate would act on.

7.1 What goes in

The city's live temperature and humidity, today's high, the 7-day forecast, the heat-stress numbers (Heat Index, wet-bulb, WBGT, work/rest category), the IMD-threshold outlook (worst level this week, any multi-day spell, the first hot day), land cover (built-up and tree share), the composite risk score, and a costing block (section 7.4). Nothing else: no ward names, no population, no facility lists, because the project holds none.

7.2 The rule-based pre-assessment

Before any AI runs, deterministic rules set an activation level, shown with its reasons:

Level	Trigger (any one)
CRITICAL	air temperature (now or today's high) ≥ 45 °C · Heat Index "Extreme Danger" · WBGT black flag (≥ 32.2 °C) · the outlook shows heatwave or severe conditions this week
ALERT	≥ 42 °C · Heat Index "Danger" · WBGT red flag · a day above the IMD threshold this week
WATCH	≥ 37 °C · Heat Index "Extreme Caution" · WBGT yellow flag
COLD	current temperature below 10 °C (a cold-weather protocol applies)
ROUTINE	none of the above

7.3 Which AI writes the plan

The server tries three tiers in order and labels the result: (1) an **Agent37**-hosted agent, if an instance is configured (the code path is built; no instance is configured in production, by the team's decision not to add a payment card); (2) **Google Gemini**, with the same prompt, in JSON mode, using the key the AGNI chat already uses; (3) the **rule-based protocol**, if both are unreachable. The panel shows a green banner naming the engine, model and time for AI output, or an amber "Offline protocol — not generated by an AI" banner with the reason. The browser side never hangs: "Use offline protocol now" is available while the AI works, and a time-out falls back automatically.

7.4 What comes out

- **Headline and situation**: one sentence, then two or three explaining the data and the level.
- **Actions**, 4–10, each with a title, concrete detail, an owner (a district role: Municipal Commissioner / ULB, CMHO / District Health, Labour Department, District Information Officer, PHED / Jal Board, DDMA control room, Fire & Emergency Services, the power distribution company, District Education Officer), a deadline and a priority P1–P3.
- **Labour order**: shift window (e.g. 06:00–11:00), restriction (12:00–16:00 no outdoor work), work/rest cycle.
- **Public advisory** as an SMS in English (kept within the GSM character set) and Hindi.
- **SDMA incident**: type, severity, summary.

- **Estimated cost** (7.5) and **If this is not controlled** (7.6).
- **Assumptions**: what the AI could not verify. In testing, Gemini listed missing items here rather than guessing, which is the behaviour the prompt demands.

7.5 Estimated cost

A "Seasonal budget to plan with" selector (₹25 L, ₹50 L, ₹1 Cr, ₹5 Cr) feeds the Mitigation Planner, and the Co-Pilot receives two costings it may allocate but not re-price:

- **This week's emergency response** at the activation level, priced from the planner's published unit costs: at CRITICAL, 4 cooling centres × ₹5 L and 10 drinking-water stations × ₹50,000 = ₹25 L; at ALERT, 2 and 6. Items with no unit cost in the model (SMS blasts, ORS stock, fuel for misting runs, overtime) are listed as "not priced", never guessed.
- **Seasonal investment**: what the planner would buy with the chosen budget (cool roofs, plantation, reflective surfaces, cooling units) and the projected composite-risk change, plus the investment needed for a 15 % risk cut, or a statement that the model's ceiling is lower.

Every figure is labelled a model-based estimate from published unit costs, not a quotation.

7.6 If this is not controlled

A red card with a summary, three to six consequences each traceable to a number on screen (days at or above the IMD threshold, the Heat Index band and its documented effect, work hours lost per worker per day from the WBGT category, wet-bulb), and who is hit first (outdoor workers, vendors and riders, the elderly and infants in poorly cooled housing, patients). Below it, a **chart**: for each of the next seven days, the hours an outdoor worker on a normal 09:00–17:00 shift spends inside the 12:00–16:00 danger window on days the forecast Heat Index reaches "Extreme Caution" or worse (4 h), against 0 h under the protocol's 06:00–11:00 order. "Show the theory" explains the dose–response reasoning (how hot × how long) with sources, and states plainly that no deaths or admissions are estimated because no mortality model is held.

7.7 Simulated actions and the Executive Brief

Three buttons, "Dispatch ward advisory", "Simulate SMS broadcast" (it counts characters and SMS segments) and "Log incident to SDMA", write time-stamped entries with reference numbers (e.g. SDMA-RJ-171020261423) to an action log kept in the browser. Nothing is sent anywhere, and the log says so. **Export Executive Heat Action Brief** opens a printable one-page document for the District Magistrate: situation, the heat-stress numbers, the 7-day forecast against the IMD threshold, the actions table, the labour order, the SMS texts, the SDMA incident, the cost, the consequences, the action log, assumptions, data sources, and which engine produced the plan.

8. The Mitigation Planner and the cost model

The planner answers "where should the money go?" for a state (across its cities) or for one city. It is deterministic: the same inputs always give the same plan, and every assumption is printed.

8.1 Three modes

Mode	You enter	You get
I have a budget	A rupee amount (presets ₹25 L to ₹5 Cr, slider to ₹10 Cr)	An optimised split across cities and intervention types, risk before → after, allocation bars, "why this plan" bullets, a city-by-city table
I want to estimate a cost	One intervention and a quantity	Cost, projected cooling, risk before → after, the calculation lines
I have a target	A desired risk reduction (2–30 %)	The investment needed, found by binary search over the optimiser, or a statement that the target is beyond the model's ceiling

8.2 The composite risk score (0–100)

```
t          = the higher of the forecast high and the live temperature
tempPts    = 10 (t < 25) · 30 (< 30) · 50 (< 35) · 70 (< 40) · 85 (< 45) · 95 (≥ 45)
surfacePts = clamp(MODIS season-peak surface − 35, 0, 10)
builtPts   = built-up % × 0.2
canopy     = clamp(tree cover % / 30, 0, 1) × 10
risk       = clamp(round(tempPts + surfacePts + builtPts − canopy), 0, 100)
```

Labels: ≥ 80 CRITICAL, ≥ 65 HIGH, ≥ 45 MODERATE, otherwise LOW. A confidence grade (high / medium / low) reflects how much of the input is real data for that city rather than a fallback.

8.3 Unit costs (all printed in the app)

Item	Unit cost	Basis
Cool roof, lime-wash, 1,000 sq ft	₹2,500 per roof	₹1.25 per sq ft (midpoint of the Ahmedabad 2017 pilot's ₹0.5–2) + ₹1,250 labour
Plantation	₹1.40 L per hectare	400 trees × ₹350 including three years' care
Cooling centre	₹5 L	assumption
Drinking-water station	₹50,000	assumption
Reflective pavement	₹15 L per hectare	₹150 per m^2
School / hospital heat retrofit	₹1.5 L	assumption

8.4 How the optimiser works

Each city's "priority core" is a 1 km radius (about 314 ha). Candidate packages are generated per city (1 ha of plantation, 100 cool roofs, a cooling centre, 1 ha of reflective surface, a facility retrofit), each with an estimated risk-point gain: cooling of 18 °C per unit of tree fraction, 14 per unit of roof fraction, 12 per unit of reflective fraction (the same illustrative coefficients the dashboard sliders use), at 4 risk points per °C; cooling centres 1.5 points, water stations 0.15, facilities 0.4. The optimiser is greedy: it repeatedly buys the package with the best points-per-rupee, weighted by the city's risk and built-up share, with diminishing returns (×0.9 for every ₹25 L already spent in the same city) and a 40 % cap on relief spending, until the budget is exhausted. The reported reduction carries a ±20 % band. Population is not modelled, because no verified per-city table is held; the planner says so.

9. The city dashboard, tab by tab

9.1 Overview (both views)

Panel	What it shows	Source
Weather Conditions	Temperature, feels-like, max/min, humidity, wind, pressure, visibility, UV, cloud, sunrise/sunset, AQI with PM2.5/PM10/NO_2/O_3, 7-day and 12-hour forecasts	Open-Meteo, live
How dangerous is the heat?	"Feels like X °C" with a danger word (Safe / Take care / Be very careful / Dangerous / Extremely dangerous), what outdoor workers should do, the worst hour of the next 24. "Show the science (for experts)" reveals Heat Index, wet-bulb, WBGT and every formula with today's numbers substituted	Formulas on live temperature + humidity
AI Heat Response Co-Pilot (authority)	Section 7	
Heatwave Action Checklist (authority)	NDMA / Ahmedabad HAP steps that adapt to severity: 5 activation steps at Extreme/High risk; 3 preparedness steps at Medium; a 4-step cold-weather list below 10 °C; "routine monitoring" otherwise. Ticks are time-stamped and kept per city; the panel calls itself "an operational aid, not an official record"	Rules
7-Day Heatwave Outlook	Seven bars of forecast maxima against the IMD threshold for the terrain, with any multi-day spell named	Open-Meteo + IMD rules
30-Day Temperature Trend	Line chart from the app's own daily archive	Archive
Satellite Indices	Modelled surface temperature, vegetation / built-up / water %, elevation	Open-Meteo, ESA, SRTM
Heat Risk Gauge, Today's High vs Low, Active Alerts, Health & Safety Precautions	Rule-based summaries	Live readings
Safe hours today, Share with family, How you can help (citizen)	A 24-hour bar coloured by feels-like (green < 35, yellow 35–39, red ≥ 40), a WhatsApp message, a static help list	Open-Meteo hourly

9.2 Prediction (authority)

The District Brief: when the heat is coming (spell dates), how bad today (composite risk), what changes if you act (risk before → after); the 7-day outlook with "if nothing is done" against "with the plan"; the checklist as a table with an owner and a due-by per step; the budget split; and two 10 × 10 grids, today versus with the plan.

9.3 Analysis (authority)

The temperature heat grid; the NASA MODIS panel (latest clear-sky day and night, hottest surface this season, weekly chart with cloud gaps, year-by-year peak-season chart back to 2016); the Random Forest model scorecard; land use / land cover; urban morphology (live building count); the data pipeline list.

> **About the machine-learning model.** It is a research prototype and the panel says so: a scikit-learn Random Forest trained on 341 rows from 20 global megacities (not India) to predict surface temperature from vegetation, built-up and elevation. On a random split it scores R^2 0.95; when five whole cities are held out it scores R^2 −0.39, meaning it does not generalise to unseen cities. The app shows both numbers and uses the model for nothing on screen. This honesty is itself a feature: the panel exists to show what was tried and why it is not trusted.

9.4 Compare (both views)

Up to four cities on a radar chart: surface temperature, AQI and wind are live; NDVI, NDBI and NDWI are illustrative baselines and the panel says so.

9.5 Interventions (authority)

A recommended plan against the 30 % canopy target; sliders for greening, cool roofs and water bodies with a before/after grid; the Cool Roof ROI calculator (lime-wash ₹0.5–2 per sq ft, premium ₹15–30; AC savings at 0.09 kWh per sq ft per month × ₹7 per kWh over eight cooling months; payback period); the physics cards; the city-level Mitigation Planner; wind and atmosphere.

9.6 AI + Export (authority)

AGNI chat; points and badges; glossary; copy summary, WhatsApp share, CSV download, printable PDF report.

10. AGNI, the AI analyst

AGNI (Analytical Ground-level heat iNtelligence Interface) is a chat assistant grounded in the numbers on screen. The browser sends the question to the app's own server, which calls Google Gemini; the API key never reaches the browser. The model is given the selected city's live readings, readings for any other city or state named in the question, an India-wide ranking block for "hottest / cleanest" questions, a satellite line, and the preferred reply language. Its instructions: use only the numbers given, say "no data" otherwise, tag anything else "(estimated)", stay on heat, climate, air quality and planning, India only, never write code, and reply in the language the user writes in (it handles Hindi and Hinglish). Limits: 10 questions a minute per visitor; if the AI is unreachable, heat questions get a templated answer from the on-screen data, clearly marked as an offline fallback. The site's interface is translated into 11 Indian languages.

11. Every formula and threshold

11.1 Heat Index (US NWS)

```
T in °F, RH in %
simple = 0.5 × (T + 61 + (T − 68) × 1.2 + RH × 0.094)
if mean(simple, T) < 80 °F → HI = simple
else (Rothfusz 1990):
HI = −42.379 + 2.04901523 T + 10.14333127 RH − 0.22475541 T·RH − 0.00683783 T² −
0.05481717 RH²
     + 0.00122874 T²·RH + 0.00085282 T·RH² − 0.00000199 T²·RH²
RH < 13 % and 80–112 °F: HI −= ((13 − RH)/4) × √((17 − |T − 95|)/17)
RH > 85 % and 80–87 °F:  HI += ((RH − 85)/10) × ((87 − T)/5)
Bands: Caution ≥ 26.7 °C · Extreme Caution ≥ 32.2 · Danger ≥ 39.4 · Extreme Danger ≥
51.7
Above 135 °F (57 °C) the NWS chart ends; the app shows "> 57 °C".
```

11.2 Wet-bulb (Stull 2011)

```
Tw = T·atan(0.151977·√(RH + 8.313659)) + atan(T + RH) − atan(RH − 1.676331)
     + 0.00391838·RH^1.5·atan(0.023101·RH) − 4.686035      (T in °C; valid RH 5–99 %,
T −20…50 °C; ±1 °C)
```

11.3 WBGT, shade estimate (ISO 7243) and work/rest (TB MED 507)

```
WBGT = 0.7 × Tnw + 0.3 × Tg,  natural wet-bulb Tnw ≈ Tw,  globe Tg ≈ air T in shade
Category 1 ≥ 25.6 °C: no limit · 2 ≥ 27.8: 50/10 min · 3 ≥ 29.4: 40/20 · 4 ≥ 31.1:
30/30 · 5 ≥ 32.2: 20/40 (moderate work)
Work hours lost per 8-h shift = rest minutes per hour / 60 × 8
```

11.4 Demo humidity (why a forced 46 °C does not give an impossible number)

When a rehearsal link forces a temperature, the app keeps the day's real moisture content (dew point) and recomputes relative humidity at the new temperature with the Magnus formula, because heating air lowers its RH. Without this, 46 °C with an evening's 60 % RH would give a 38 °C wet-bulb, which has never been observed on Earth.

11.5 State colour

Take every city in the state with a reading; the state's temperature is their median; the colour is the bucket with the most cities, ties to the hotter bucket. Readings under 45 minutes old are used alone when at least min(5, n) exist. Under the spectral palette the colour is the continuous colour of the median.

11.6 IMD heatwave outlook

```
threshold = 40 °C plains · 37 °C coast · 30 °C hills (elevation ≥ 900 m)
day level: ≥ 47 severe · ≥ 45 heatwave · ≥ threshold watch · ≥ threshold − 3 warm ·
```

```
else normal
spell = 2+ consecutive days at watch or worse
```

11.7 Urban heat island gap

```
UHI = city temperature − temperature at (lat + 0.35°, lon + 0.10°)   ≤ 0 parity · ≤
1.5 mild · ≤ 3.5 moderate · > 3.5 severe
```

11.8 Interpolation (IDW)

```
value(x) = Σ wᵢ·vᵢ / Σ wᵢ,  wᵢ = 1 / (dᵢ² + ε), over the 8 nearest readings; blank
beyond 1.5° (~150 km), 4° on sparse layers
district value = direct city reading if names match, else 1/(km+5)-weighted mean of
the 2 nearest same-state cities within 150 km
```

11.9 Wind vectors

```
Open-Meteo gives speed (km/h) and direction the wind blows FROM (°). u =
−speed·sin(dir), v = −speed·cos(dir)
```

11.10 Air quality bands (US AQI)

≤ 50 Good · ≤ 100 Moderate · ≤ 150 Unhealthy for Sensitive Groups · ≤ 200 Unhealthy · ≤ 300 Very Unhealthy · > 300 Hazardous.

12. Real, estimated, or fixed? The honesty rule

Measured or live

- City temperature, humidity, wind, forecast, AQI and pollutants (Open-Meteo); state medians and colours; the hottest-cities list.
- NASA MODIS surface temperature, quality-filtered; cloud days left as gaps.
- NASA VIIRS daily imagery; Esri imagery.
- ESA WorldCover land cover for 171 cities; OpenStreetMap building counts; SRTM elevation.

Estimated, and labelled as such on screen

- A state with no live cities ("estimated"); land cover for a city outside the 171 ("est. from X, ~N km"); districts without their own city ("nearest cities …" or "estimated (rounded) from …").
- The modelled "Surface Temp (live)" on the Overview, labelled as not a satellite reading.
- The heat grid's ±2 °C pattern, the intervention sliders, and every planner outcome ("model-based estimates").
- The WBGT ("shade estimate"), the wind probe's heat-load and ventilation labels ("heuristic"), the 14-day satellite composite.
- Demo-link temperatures (banner: "this browser only, not real data").

Fixed text with no live source (candidates for removal before judging)

- The Global Climate Systems panel (ENSO "El Niño", IOD, MJO, PDO, NAO, Polar Vortex), the ticker items "El Niño Active" and "Sat Pass: Next 16:45 IST", the pills HEAT HIGH / CLIMATE EL NIÑO / SYSTEM OPS, the "AGENT ONLINE" and "SAT ACTIVE" badges, the "Marine heatwave active" alert for four coastal states, and the progress tracker.

13. Technical architecture

Front end	React 18 + Vite; Tailwind and plain CSS; Leaflet for the Command Center map (no API key, no paid service); react-simple-maps + d3 for the classic map; Recharts for charts; HTML5 canvas for the wind particles and the thermal raster; i18next for 11 languages.
Server	Vercel serverless functions: `/api/ask-ai` (AGNI → Gemini), `/api/heat-agent` (Co-Pilot: Agent37 → Gemini → rules), `/api/weather-history`, and two cron routes for the cache refresh. The Vite dev server mirrors them, so `npm run dev` runs everything locally.
Secrets	Only server-side environment variables: `GEMINI_API_KEY`; optional `AGENT37_API_KEY` / `AGENT37_INSTANCE_ID`; `GITHUB_TOKEN` and `CRON_SECRET` for the refresh. No key is ever shipped to the browser. All map and weather services used by the browser need no key at all.
Data refresh	GitHub Actions (hourly by day, three-hourly at night) commits the bulk cache and a daily snapshot; the commit redeploys the site.
Offline behaviour	Map, cached readings, satellite archive, land cover, planner, heat stress, the Co-Pilot's rule-based protocol, the brief and exports all work without internet. Live fetches, AI calls, imagery tiles and the wind layer need it; each fails to a labelled fallback, never to a blank screen (error boundaries wrap the app and the map).
Validation	Every AI reply is parsed, validated and clamped before it reaches the page; URL parameters are strictly parsed (junk values are ignored, never coerced); rate limits protect the AI routes.
Storage	Accounts, settings, checklist ticks, action logs and last-known weather live in the browser's local storage. Nothing about users is sent to a server.

14. Limits and known gaps

- No ward boundaries, population, or facility locations are held, so plans describe zones by type and the AI is barred from inventing wards. Deep zoom shows real roads and buildings from Esri/OpenStreetMap, not wards.
- No mortality model: the app never estimates deaths or hospital admissions.
- Satellite surface temperature is the ground, not the air; the 14-day composite is labelled; monsoon weeks have few clear-sky readings.
- Land cover covers 171 cities (2021 data); schools and hospitals only 6 cities.
- Accounts are browser-only; the inner sign-in screen after logout accepts any non-empty credentials.
- Agent37 is integrated but not active in production (no instance), so the Co-Pilot runs on Gemini; the team cannot claim Agent37 use.
- Several fixed-text widgets remain (section 12) and should be labelled or removed before judging.

- Open-Meteo's free tier can rate-limit a single IP after very heavy use; the app then shows cached readings, labelled.

15. Demo script and judge questions

A three-minute walkthrough

1. **0:00–0:40, the problem.** Open https://heatops.vercel.app, sign in. "India loses thousands of lives to heat every year; a district officer has an IMD bulletin and a phone. This is the All-India view: 1,932 cities live, satellite surface temperature, and the hottest city right now."
2. **0:40–1:20, the data.** Switch to Satellite, then Surface temperature; drag Historical to 2019, then 2024: "real NASA readings, year by year". Switch to Live Pass with NASA true colour: "the actual photograph of that day". Zoom into a state: per-district colours; hover a district: "reading from Unnao".
3. **1:20–2:20, operations.** Open the rehearsal link (`?demoTemp=46`) and pick a city: the triage turns CRITICAL. Run the AI Co-Pilot: actions with owners and deadlines, the labour order, the SMS in Hindi, the cost in rupees from published unit costs, and the red "if this is not controlled" chart. Click Simulate SMS and Log to SDMA; export the brief.
4. **2:20–3:00, defence.** "Every number has its source next to it. Show the science: the Heat Index formula with today's values. The planner's unit costs come from the Ahmedabad pilot. What we don't have, we say: no ward data, no death model. Open-source maps, no paid keys; it runs on a laptop without Wi-Fi except for the live calls."

Questions to expect, and the honest answers

Question	Answer
Is this just a weather wrapper?	The weather is one input. The product is the triage rules, the heat-stress maths, the satellite archive, the cost model, the optimiser and the AI protocol, all on screen with their working.
Why is the AI output trustworthy?	It is given only our numbers and unit costs, told to list what it lacks, and its reply is validated before display. The banner names the engine. The rule-based protocol is one click away and is what runs offline.
Why are the hills red in the satellite view?	Because MODIS measures the ground at 10:30 am; bare high-altitude rock gets hotter than the air. The tooltip says "surface", and the long-record cities in J&K are on the Jammu side.
Where are the ward boundaries?	There is no public dataset for them; we show real roads and buildings and never label anything a ward.
How many deaths will this prevent?	We don't estimate deaths; no validated India-wide model exists. We show exposure hours and work hours lost, which are measurable, and cite the Ahmedabad plan's published effect.
Can a government deploy this?	Yes: open-source stack, no paid keys, server-side secrets, IMD or CPCB feeds can replace Open-Meteo behind the same interface; Open-Meteo itself can be self-hosted for a paid deployment.

16. Glossary

Term	Meaning
AQI	Air Quality Index, 0–500, from pollutant concentrations; the app uses the US EPA scale.
Built-up fraction	Share of land covered by man-made surfaces (ESA WorldCover class 50).
Choropleth	A map that colours areas by a value.
Composite	A satellite layer built from each location's most recent clear-sky reading within a window.
Dew point	The temperature at which the air's moisture would condense; a measure of absolute humidity.
Elevation	Height above sea level, metres.
Heat Action Plan (HAP)	A city or district's pre-agreed set of heatwave measures (Ahmedabad's is the Indian model).
Heat Index	NWS "feels like" temperature from air temperature and humidity.
IDW	Inverse-distance weighting: interpolation that weights nearer readings more.
IMD	India Meteorological Department, the official forecaster and issuer of heatwave warnings.
LST	Land surface temperature, the ground's temperature as seen by a satellite.
MODIS	NASA's Moderate Resolution Imaging Spectroradiometer on the Terra and Aqua satellites; daily 1 km LST.
NDMA	National Disaster Management Authority, author of India's heatwave guidelines.
Open-Meteo	A free weather API aggregating national weather models; no key required.
PM2.5 / PM10	Airborne particles under 2.5 / 10 micrometres.
SDMA / DDMA	State / District Disaster Management Authority.
SRTM	NASA's Shuttle Radar Topography Mission, the global elevation survey.
UHI	Urban heat island: a city warmer than its surroundings.
ULB	Urban local body (municipal corporation or council).
VIIRS	NASA/NOAA imager on the Suomi NPP satellite; the daily true-colour picture.
WBGT	Wet-Bulb Globe Temperature, the occupational heat-stress standard (ISO 7243).
Wet-bulb temperature	The lowest temperature reachable by evaporative cooling; the physical limit of sweating.

Sources named in the app: NWS Heat Index (Rothfusz 1990); Stull, J. Appl. Meteor. Climatol. 50 (2011); ISO 7243:2017; US Army TB MED 507; Vecellio et al., J. Appl. Physiol. 132 (2022); IMD heatwave criteria; NDMA heat-wave guidelines; Ahmedabad Heat Action Plan; NASA LP DAAC MOD11A1.061; ESA WorldCover 2021; NASA GIBS; Open-Meteo.